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28 JULY 1954
SUPERSEDING
MIL-A-20547
29 DECEMBER 1951

MILITARY SPECIFICATION

AMPOULE, GLASS, SOLVENT FILLED

This specification has been approved by the Department of Defense for use of the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers solvent-filled glass ampoules for use in bomb fuzes.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-G-2550 — General Specification for Ammunition Except Small Arms Ammunition.

STANDARDS

MILITARY

MIL-STD-129— Marking for Shipment and Storage.

DRAWINGS

ORDNANCE CORPS

73-8-174 — Fuze, Bomb, Tail, M123A1, M124A1, and M125A1. Details

PUBLICATIONS

ORDNANCE CORPS

ORD-SIP-M6-388—Ampoule for Fuzes, Bomb, Tail, M123A1, M124A1, and M125A1.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.—Material and parts shall be in accordance with applicable specifications and drawings.

3.2 The ampoule shall comply with all requirements specified on Drawing 73-8-174, and all drawings referenced thereon, and with all requirements specified in applicable specifications.

3.3 The unfilled ampoules shall show no indication of strain as indicated by a milky area in the glass when tested as specified in 4.4.

3.4 The ampoules shall be filled as specified in the contract.

3.5 Filled ampoules.

3.5.1 The filled ampoules shall show no indication of strain (indicated by a milky area in the glass) except in the constricted end produced by sealing, when tested as specified in 4.6.1. Strain in the constricted end shall be allowed provided it is not greater than that shown by samples 5A and 5B representative of permissible defects (see 6.1).

3.5.2 *Nonbreaking impact.*—The ampoules shall not break under the impact of a standard $^{15}/_{16}$-inch-diameter steel ball weighing 1.93 ounces dropped from a height of 5 inches when tested as specified in 4.6.2.

3.5.3 *Breaking.*—The ampoules shall break under a load of 475 pounds maximum when tested as specified in 4.6.3.

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3.6 When tested under the temperature conditions specified in 4.7, the ampoules shall comply with the following requirements:

3.6.1 *Cracks.*—The ampoules shall have no cracks.

3.6.2 *Leakage.*—Each ampoule shall be at least ⅔ full and shall show no evidence of leakage.

3.6.3 *Sharp edges.* — The ampoules shall have no defects which feel sharp to the touch such as chip-outs or broken-out bubbles as shown by samples 7A and 7B, respectively, representative of nonpermissible defects (see 6.1).

3.6.4 *Folds.*—The ampoules shall contain no folds such as shown by samples 3A and 3B representative of nonpermissible defects (see 6.1).

3.6.5 *Bubbles.* — The ampoules shall contain no large bubbles such as shown by samples 9A, B, and C nor excessive small bubbles such as shown by samples 8F, G, H, I, and J representative of nonpermissible defects (see 6.1). Several small separated bubbles shall be allowed provided they do not exceed those shown by samples 8A, B, C, D, and E representative of permissible defects (see 6.1).

3.6.6 *Indentations.*—The tip of the ampoules shall have no indentations such as shown on samples 4A and 4B, representative of nonpermissible defects, and shall have a contour as shown by sample 1A (see 6.1). Slight smooth indentations shall be permitted in other parts of the ampoule provided they do not exceed those shown by sample 2B representative of permissible defects. Ampoules shall contain no deep indentations such as shown by sample 2A representative of nonpermissible defects.

3.6.7 *Scratches.*—Surface scratches (such as caused by tools, fixtures, etc.) shall be permitted provided they are not deeper than those shown by samples 6A and 6B representative of permissible defects (see 6.1).

3.7 Workmanship.—The ampoules shall be homogeneous, transparent, practically colorless, and free from mechanical imperfections. All parts shall be free of chips, dirt, grease, and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot.—A lot shall consist of ampoules of one type loaded and assembled by one manufacturer under one contract in one unchanged process in accordance with the same drawing, same drawing revision, the same specification, and the same specification revision. Each lot shall contain:

(a) acetone from not more than two lots.

(b) alcohol from not more than two lots.

4.2 Sampling.—Number of samples and acceptance and rejection criteria used for determination of lot acceptance for tests specified in 4.5 and 4.6.3 shall be in accordance with Standard Inspection Procedure ORD–SIP–M6–388.

4.3 Inspection.—Inspection shall be as specified in Specification MIL–G–2550.

4.4 Detection of strains in empty ampoules.—Each empty ampoule shall be examined by means of an AO Polaroid Colmascope Model No. M–64, or approved substantial equal, for evidence of strains to determine compliance with 3.3.

4.5 Test of acetone-alcohol mixture. — Sample ampoules shall be selected, and the percentage of acetone and alcohol in the sample shall be determined as follows:

(a) A pipette shall be constructed with a capillary tip of approximately 1-ml capacity by drawing out the point of a 1-ml pipette. The bore of the pipette shall be drawn out above the bulge to a thin tube and marked at

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the thinnest point, so as to obtain a reproducible volume of liquid.

(b) Fifteen grams of hydroxylamine hydrochloride shall be dissolved in 1500 ml of water, and the solution brought to a pH of 4.35 with approximately third-normal hydrochloride solution in an Erlenmeyer flask. The tip of the constricted end of the ampoule shall be filed with emery paper until an opening appears big enough to allow insertion of the capillary tip of the pipette. The pipette shall be filled with the solution in the ampoule and emptied under the surface of the hydroxylamine-hydrochloride solution by blowing. The Erlenmeyer flask shall be stoppered immediately, shaken thoroughly, and allowed to stand for ½ hour. The contents of the flask shall be washed into a 400-ml beaker, the solution diluted to 200 ml, and titrated electrometrically with third-normal sodium hydroxide solution. The pH shall be brought to 4.35. A glass electrode shall be used in making this determination.

(c) The third-normal sodium-hydroxide solution shall be standardized with pure acetone using the same pipette and the same procedure as was used in the determination. The percentage by volume of acetone in the acetone-alcohol mixture shall be calculated as follows:

$$\text{Percent acetone (by volume)} = \frac{100A}{B}$$

where:

A = ml third-normal sodium hydroxide solution used in the determination.

B = ml third-normal sodium hydroxide solution used in the standardization.

(d) The percentage of alcohol shall be calculated by subtracting from 100 the percentage of acetone to determine compliance with the requirements specified on the drawing.

4.6 Test of filled ampoules.

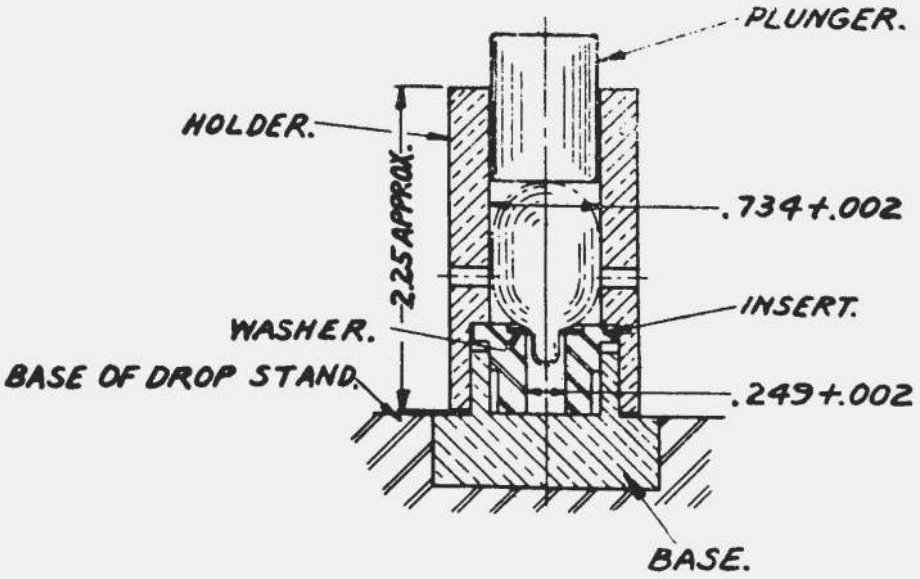
4.6.1 *Detection of strains.*—Each ampoule shall be examined for strains using an AO Polaroid Colmascope Model No. M–64, or approved substantial equal, to determine compliance with 3.5.1. If any ampoule fails to comply with 3.5.1, it shall be rejected and destroyed.

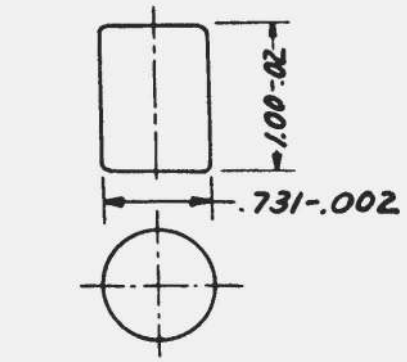
4.6.2 *Nonbreaking impact test.* — Each filled ampoule shall be tested by mounting it in the fixture shown on figure 1. A $1\frac{5}{16}$-inch-diameter steel ball, weighing 1.93 ounces shall be dropped freely through a height of 5 inches, to impact on the plunger. The aluminum washer shown on figure 1 shall be replaced after each drop. Each ampoule shall be observed for evidence of breakage to determine compliance with 3.5.2.

4.6.3 *Breaking test.* — Sample ampoules shall be assembled to ampoule caps and tested by mounting in a fixture as shown on figure 2. An increasing load shall be applied to the arming rod at a rate of approximately .0125 inch per minute. The load at which each sample ampoule assembly breaks shall be recorded to determine compliance with 3.5.3.

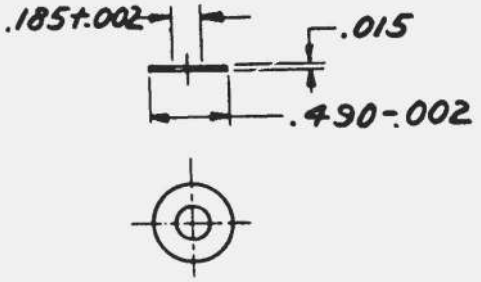
4.7 Test for visible defects.—Each filled ampoule shall be conditioned at a temperature of —50° to —65° F., for a period of 3 hours. The ampoules shall be allowed to return to room temperature (65° to 80°F.), and then immersed in boiling water (206° to 213°F.) for a period of 7 hours. Upon removal, the ampoules shall be allowed to return to room temperature in a period not less than 1 hour. Each ampoule shall then be examined under a strong light by means of a stereomicroscope of not less than 6 nor more than 7 power. Observation shall be made for the presence of visible defects to determine compliance with 3.6. If any ampoule fails to comply with 3.6, it shall be rejected and destroyed.

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PLUNGER
TOOL STEEL-HARDEN.
FINISH ∫"
1-REQ'D



WASHER
ALUMINUM, TYPE 2, CLASS "A";
SPEC. QQ-A-561
1-REQ'D

FIGURE 1.

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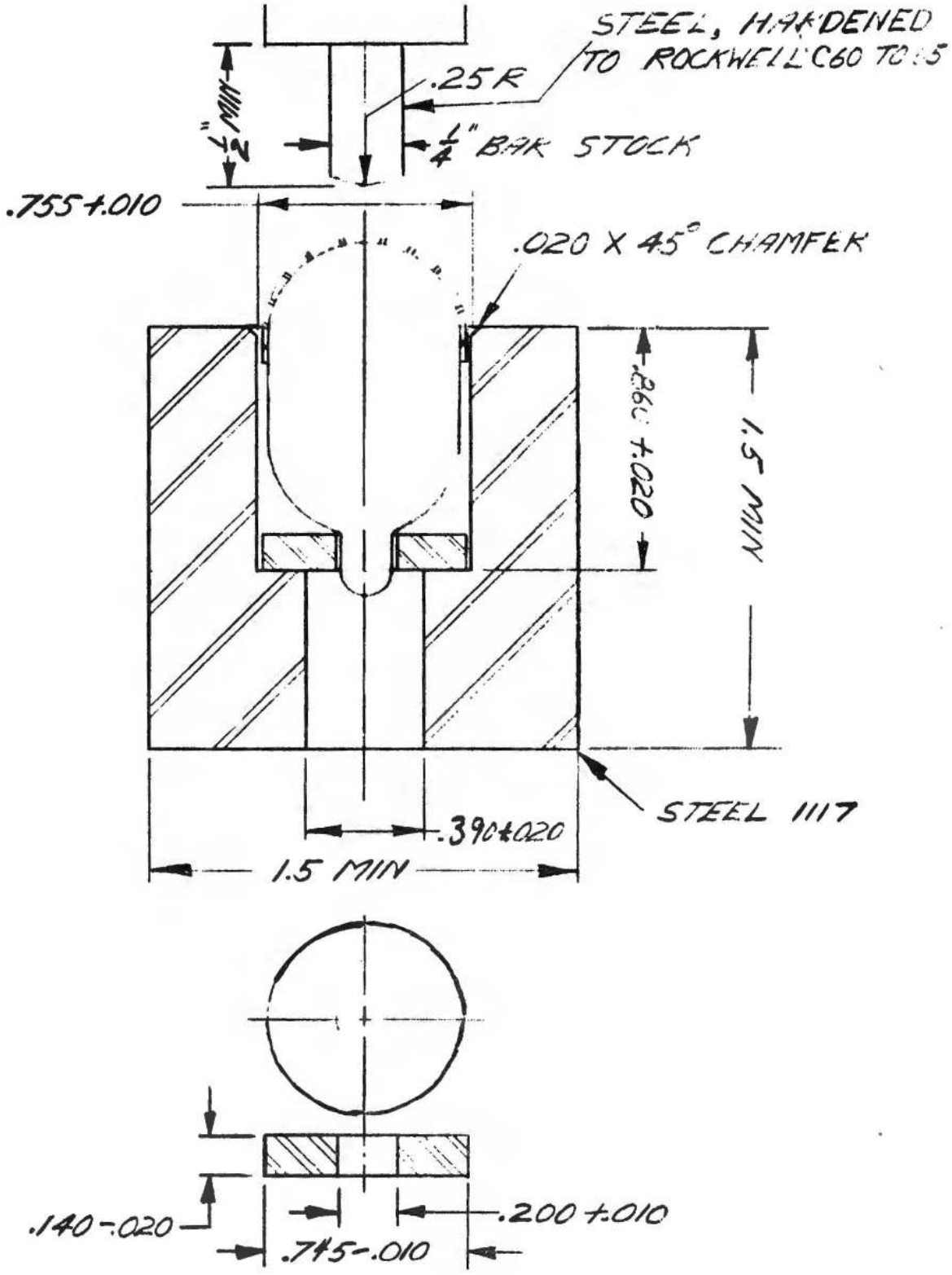


FIGURE 2. *Steel, hardened to Rockwell C60 to 65.*

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5. PREPARATION FOR DELIVERY

5.1 Packing.—Unless otherwise specified, the ampoules shall be delivered in standard commercial containers so constructed as to insure acceptance by common or other carrier, for safe transportation, at the lowest rate, to the point of delivery. The packing container shall be sufficiently rigid to withstand transportation and permit storing in tiers at least ten feet high without damage to the container or contents. The gross weight of each container and the parts to be packed therein shall be as specified in the contract.

5.2 Marking.—Marking for shipment shall be in accordance with Standard MIL–STD–129.

5.3 Data cards.—Data card information shall be as specified in Specification MIL–G–2550.

6. NOTES

6.1 Representative samples illustrating visible defects are available at Picatinny Arsenal for the information of the inspector and contractor.

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